Jimena Garcia-Nuno

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Objective

Electrical Engineering undergraduate at Georgia Tech seeking a Summer 2026 internship. Experience in embedded systems, PCB design, hardware validation, and hardware–software integration, with hands-on software development for system control and data acquisition, focused on building reliable, scalable engineering solutions.

Education

Georgia Institute of Technology | Atlanta, GA *August 2025 – Present*
Bachelor of Science in Electrical Engineering Expected Graduation, May 2029

Skills

Programming: MATLAB, Python, C++, HTML
Hardware: Microcontrollers (Arduino, ESP32, Boron 404x), Soldering station, Multimeter, PCB fabrication
Software: Autodesk EAGLE, MATLAB/Simulink, KiCAD EDA, SolidWorks, VSCode, Arduino IDE, Image J
Professional Organizations: Society of Hispanic Professional Engineers, ASHRAE, Optica
Communication: Technical reports, Lab reports, Presentations
Languages: Spanish (native), English (fluent), French (beginner)

Experience

Georgia Institute of Technology | Atlanta, GA *June 2024 – Present*
Project ENGAGES Research Student and Undergraduate Research Student, Optical Imaging and Spectroscopy Lab
Optical imaging research lab developing advanced, label-free technologies for biomedical applications including cancer detection and tissue characterization.

- Designed and fabricated custom printed circuit boards (PCBs) for quantitative oblique back-illumination microscopy (qOBM) and multi-channel imaging systems
- Developed a multi-channel handheld qOBM probe prototype using SolidWorks
- Programmed PCBs with Arduino to coordinate LED control and image acquisition within a MATLAB-based application
- Expanded system capabilities by developing fluorescence imaging functionality using multi-wavelength LED excitation
- Presented technical progress and experimental results in lab meetings and to program directors
- Implementing parallel image acquisition and saving in MATLAB to improve system efficiency by reducing RAM usage and saving data in a timely manner

Georgia Institute of Technology – Aerospace System Design Laboratory (ASDL) | Atlanta, GA *July 2025 –Present*
Undergraduate Research Assistant, IoT Lab
A systems engineering research lab developing scalable IoT sensor platforms for environmental monitoring and smart infrastructure, integrating PCB design, embedded systems, and hardware validation for field deployment.

- Redesigned PCBs for a multi-node IoT sensor system, including the HUB, Indoor Air Quality (IAQ), and Current Transformer (CT) boards, improving reliability and manufacturability
- Transitioned the lab's PCB design workflow from KiCad to Autodesk Eagle to enable standardized libraries and improve collaboration
- Created custom Eagle footprints and libraries for unavailable components, including sensors and connectors, verifying pad sizes, drill diameters, and alignments
- Re-routed boards with proper trace clearances, added vias to eliminate jumper wires, and corrected inaccurate footprints from prior generations
- Validated PCB functionality through continuity testing and prototype soldering; adjusted mechanical features such as mounting holes and drill sizes to ensure proper enclosure fit
- Improved mechanical stability and field deployment readiness by incorporating mounting and throughput holes and optimizing board geometry

Activities

FIRST Robotics Competition (FRC) Team 6023| Technical Mentor *December 2025 – Present*

- Mentor high school students in system design, wiring, and troubleshooting for competition robots
- Guide students through hands-on problem solving, safe tool use, and engineering best practices

Awards

- Amazon Future Engineer *April 2025*
- AHRAE Region IV Student Summit Competition Runner-Up Winner *October 2025*